25901384 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCMAT200 - MATRIX ALGEBRA AND NUMBER THEORY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. Compute 3A, -5A and 0A for $A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$. [U] / [CO1]

2. Show that the zero matrices are the only square matrices that are both symmetric and skew-symmetric. [U] / [CO1]

3. Define linear independence of rows of a matrix. [K] / [CO4]

4. Determine whether the matrices $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ are equivalent. [U] / [CO4]

5. Find gcd(100, 25) using Euclidean Algorithm. [K] / [CO5]

6. Find three prime numbers of the form $2^n + 1$. [U] / [CO5]

7. What is meant by a complete system of residues modulo n. [U] / [CO5]

8. Evaluate $3^{\phi(10)} \pmod{10}$. [A] / [CO8]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. a) Express the system $y + 4z = 1, x - 2y + z = 0, 3y - 4z = 23$ in matrix form and write the augmented matrix. [U] / [CO2]

   b) Explain the effect of left multiplication by the matrix $P = \begin{bmatrix} 0 & \alpha & 1 \\ 0 & \beta & 0 \\ 1 & \gamma & 0 \end{bmatrix}$.

10. Reduce the matrix $\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 2 \\ 3 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 & 2 \end{bmatrix}$ to row-echelon form. [A] / [CO3]

11. For each integer n, let $A_n = \begin{bmatrix} 1-n & -n \\ n & 1+n \end{bmatrix}$. Prove that $A_n A_m = A_{n+m}$. [A] / [CO4]

12. Compare the row ranks of the matrices A = $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$ and B = $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. [An] / [CO4]

13. Show that the product of four consecutive integers is 1 less than a perfect square. [U] / [CO5]

14. Show that there exist integers x and y for which $c = ax + by$ if and only if $gcd(a, b)|c$. [U] / [CO5]

15. Let $7 \nmid a$. Then show that $a^3 + 1 \; or \; a^3 - 1$ is divisible by 7. [An] / [CO6]

16. Find the remainder when 2(26!) is divided by 29. [U] / [CO7]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. Show that for every matrix A, there exists (a) an additive identity (b) an additive inverse. [U] / [CO1]

18. For what values of λ does the system of equations $x + y + 3z = 0,\ 2x + 3y + \lambda z = 0,\ -3x - 4y + z = 0$ has a non-trivial solution. Hence, find the solution. [A] / [CO4]

19. Determine for which values of α the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & \alpha \\ 0 & 1 & 1 \end{bmatrix}$ is invertible. For those values, find A^{-1}. [An] / [CO4]

20. (a) State Division algorithm. [U] / [CO5]
(b) Write -25 in the form $7q + r,\ 0 \leq r < 7.$
(c) Show that the expression $\frac{a(a^2+2)}{3}$ is an integer for all $a \geq 1.$

21. (a) Show that every composite number $F_n = 2^{2^n} + 1$ is a pseudoprime, where $n = 0, 1, 2, \ldots$ [An] / [CO6]
(b) Show that 341 is pseudoprime.

22. (a) Define Euler's phi function. [U] / [CO8]
(b) Show that if $n > 2$, $\phi(n)$ is even.
(c) Evaluate $\phi(840)$.

[10x3 = 30]